FRAMINGHAM HEART STUDY

Heart Health Risk Analysis

Predictive modeling, clinical decision threshold calibration, and patient phenotype clustering.

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DATA HYGIENE & ARCHITECTURE

Cohort Dimensions & Preprocessing

4,240
Total Patients

15.2%
10-Year CHD Event Rate

16
Clinical Features

388
Missing Glucose Values

Pipeline Imputation

Median imputation inside Scikit-Learn pipelines prevents data leakage.

Stratified Sampling

80/20 train-test partition preserves the 15.2% minority class balance.

STAGE 1: CONTINUOUS PREDICTION

Systolic Blood Pressure Regression

Model Fit vs Baseline

0.230

Test R^2 (Mean Baseline: -0.001)

18.47 mmHg

Root Mean Squared Error (RMSE)

Key Feature Associations

Body Mass Index (BMI): +1.497 mmHg per unit

Age: +0.871 mmHg per year

Glucose: +0.068 mmHg per unit

Male Sex: -3.000 mmHg offset

STAGE 2: STANDARDIZED CLASSIFICATION

10-Year CHD Risk Multipliers

Biomarker	Std. Coefficient	Odds Ratio (OR)	Clinical Interpretation
Age	+0.577	**1.781**	Primary long-term risk multiplier
Systolic BP	+0.302	**1.353**	Direct hemodynamic vascular stress
Cigarettes Per Day	+0.271	**1.311**	Dosage-dependent endothelial damage
Male Sex	+0.219	**1.245**	Elevated baseline coronary prevalence

STAGE 3: DECISION CALIBRATION

The Clinical Accuracy Paradox

Threshold	True Positives	False Negatives	Sensitivity	Specificity	Accuracy
0.50 (Default)	7	**122**	**5.4%**	98.6%	84.4%
0.30 (Balanced)	34	95	26.4%	91.0%	81.1%
0.15 (Screening)	**80**	**49**	**62.0%**	67.9%	67.0%

Clinical Takeaway: Lowering the cutoff to 0.15 boosts sensitivity from 5.4% to 62.0%, catching 73 more cardiac patients.

DISCRIMINATION & EXTERNAL VALIDITY

ROC Performance & Transferability

0.702

Test ROC AUC

Meaningful discriminatory baseline above random ranking (0.50).

External Validation Constraints

- **Historical Cohort: Framingham data reflects historical demographics.**
- **Demographic Drift: Model weights cannot transfer directly to target clinics.**
- **Local Recalibration: Essential before deploying into routine clinical workflows.**

STAGE 4: UNSUPERVISED PHENOTYPING

Four Patient Risk Archetypes

Cluster 1: Optimal

6.5% CHD

Youngest normotensive cohort (Age 42.6, BP 118.6).

Cluster 2: Aging

17.6% CHD

Older cohort with hypercholesterolemia (Chol 260.3).

Cluster 0: Hyper

24.7% CHD

Elevated BMI & severe vascular hypertension (BP 159.6).

Cluster 3: Metabolic

51.2% CHD

Severe diabetic subgroup with glucose > 250 mg/dL.

STAGE 5: SURROGATE EXPLAINABILITY

Decision Tree Extraction (83.3% Fidelity)

Interpretable Rules

A depth-4 decision tree translates multi-dimensional k-means clusters into transparent clinical criteria, isolating metabolic outliers and hypertensive risk stages.

Primary Diagnostic Nodes

- **Root Node: Age cutoff isolates the younger normotensive group**
- **Hemodynamic Node: Systolic BP > 141.5 splits severe hypertension**
- **Metabolic Node: Fasting glucose > 170.5 separates Cluster 3**

CONCLUSIONS & NEXT STEPS

Synthesis & Recommendations

Screening Thresholds

Adopt low thresholds (0.15) for clinical screening rather than standard 0.50 cutoffs.

Phenotype Triage

Deploy auditable rule trees to prioritize interventions for high-risk metabolic clusters.

Local Calibration

Perform prospective validation before real-world patient deployment.